

Antenna engineering — the working basics

Companion notes for the 2.45 GHz pyramidal horn in this repo · print and write on

An antenna is an impedance transformer between a transmission line and free space, and every quantity below is a way of asking how well it does that, or where the energy goes once it has. This covers what you need to read an HFSS result and know whether to believe it. Symbols follow Balanis.

λ	free-space wavelength, c/f. At 2.45 GHz, 122.4 mm
a, b	waveguide broad and narrow inside dimensions
a_1, b_1	horn aperture, H-plane and E-plane
ρ_1, ρ_2	E- and H-plane flare radii (apex distances)
Γ	reflection coefficient at the feed; S_{11} is the same thing in dB

1 · What the antenna is doing

A transmission line carries a guided wave: the fields are bound to the conductors and nothing radiates. Free space carries an unguided wave with a characteristic impedance of $\eta_0 = 376.7 \Omega$. An antenna is the structure that gets you from one to the other without reflecting most of the power back where it came from. A horn does it gradually — that is the whole idea of the flare.

Reciprocity. The pattern, gain and impedance of a passive antenna are identical transmitting and receiving. This is why the radar can use two identical horns and why you may simulate only the transmit case. It stops being true the moment anything non-reciprocal is in the path — a circulator, a ferrite, an amplifier.

Field regions

The pattern is only a pattern once you are far enough away. Three regions, with D the largest aperture dimension:

region	extent	what happens there
Reactive near field	$r < 0.62 \cdot \sqrt{(D^3/\lambda)}$	Stored energy, not radiation. Fields are largely out of phase with each other.
Radiating near field (Fresnel)	$0.62 \cdot \sqrt{(D^3/\lambda)} < r < 2D^2/\lambda$	It radiates, but the pattern still changes shape with distance.
Far field (Fraunhofer)	$r > 2D^2/\lambda$	Pattern shape is fixed; only amplitude falls, as $1/r$. This is where gain is defined.

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Why this bites you on the bench

For this horn D is the 327 mm aperture diagonal, so the far field starts at $2D^2/\lambda = 1.75 \text{ m}$. A gain measurement at 1 m is measuring the wrong thing. The two-antenna method in the build notes uses 3 m for exactly this reason. HFSS reports far-field quantities by extrapolation, so it does not care — but your tape measure does.

2 · The quantities, and what each one hides

Radiation pattern

Power density as a function of direction, normalised and plotted in dB. The features worth naming: the **main lobe**, the **half-power beamwidth** (HPBW, the angle between the -3 dB points), the **first null beamwidth**, the **sidelobes** and the **back lobe**. A uniformly illuminated rectangular aperture has a $\sin(x)/x$ pattern and a first sidelobe at -13.2 dB ; anything that tapers the illumination toward the edges pushes sidelobes down and widens the main lobe. You cannot have both.

Directivity, gain, and efficiency — three different things

term	definition	what it excludes
Directivity D	How much of the radiated power goes in the best direction, vs an isotropic radiator.	All losses. A perfectly lossy antenna can have high directivity.
Gain G	$G = e_{cd} \cdot D$, where e_{cd} is conduction and dielectric efficiency.	Mismatch. Power reflected at the port is not counted.
Realized gain	$G_{re} = (1 - \Gamma ^2) \cdot G$	Nothing. This is the number that matters , and what HFSS plots by default.

For a well-made horn the difference between D and G is a few tenths of a dB — copper loss at 2.4 GHz is small. The difference between G and realized gain is entirely about your feed match, which is the part you actually have to tune.

dB_i vs dB_d. dB_i is referenced to an isotropic radiator, dB_d to a half-wave dipole. $\text{dBd} + 2.15 = \text{dBi}$. Datasheets that quote a suspiciously good number are often quietly using dB_d, or quoting directivity, or both.

EIRP

EIRP = transmit power + antenna gain, and it is what regulators limit. This radar puts about +10 dBm into a 13.4 dBi horn, so **EIRP $\approx$ +23.4 dBm = 0.2 W** — inside Part 15 for the 2.4 GHz ISM band. Note that gain cuts both ways: a higher-gain horn is not free, it spends your legal budget.

Aperture, and the bound you cannot beat

For an aperture antenna of physical area A:

$$G = (4\pi/\lambda^2) \cdot A \cdot e_{ap} \text{ with } e_{ap} \leq 1$$

The **aperture efficiency** e_{ap} collects everything that stops the aperture radiating as if it were uniformly illuminated and perfectly in phase: taper, phase error across the mouth, spillover, blockage. An optimum pyramidal

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horn runs at about **0.51**, and that is not a defect — it is the price of the phase curvature that comes with a short flare (§ 4).

The cheapest sanity check in antenna work

$4\pi A/\lambda^2$ is the gain of a perfectly illuminated aperture of that size. **No passive antenna of that aperture can exceed it.** A solver that hands you a gain above the bound is telling you the model is wrong — usually a port that is injecting more power than you think, or a radiation boundary so close it is acting as a reflector. For this horn the bound is **16.3 dBi** and the design sits at 13.4 dBi. Compute it before you trust any simulated gain, every time.

Impedance, match and bandwidth

At the feed the antenna looks like an impedance Z_{in} . Referenced to the line impedance Z_0 (50 Ω here) it becomes the reflection coefficient:

$$\Gamma = (Z_{in} - Z_0) / (Z_{in} + Z_0) \quad S_{11} \text{ (dB)} = 20 \cdot \log_{10} |\Gamma| \quad VSWR = (1 + |\Gamma|) / (1 - |\Gamma|)$$

S_{11}	VSWR	$ \Gamma $	power reflected	mismatch loss
-6 dB	3.0	0.50	25 %	1.25 dB
-10 dB	1.92	0.32	10 %	0.46 dB
-14 dB	1.50	0.20	4 %	0.18 dB
-20 dB	1.22	0.10	1 %	0.04 dB

-10 dB is the usual pass mark, and the table shows why it is a reasonable place to stop: the last 4 % of power costs a lot of tuning. **Bandwidth** is simply the frequency range over which you still meet it. This radar sweeps 2400–2483.5 MHz, which is 3.4 % fractional bandwidth — trivially easy for a waveguide horn, whose limit is the single-mode band of the guide, not the aperture.

Polarisation

The orientation of the E-field. A rectangular waveguide in TE_{10} gives linear polarisation, parallel to the **narrow** wall (b). Two linearly polarised antennas misaligned by angle ψ suffer a **polarisation loss factor of $\cos^2\psi$** : at 45° that is 3 dB, at 90° it is in principle infinite and in practice 20–30 dB of cross-polarisation. This is why the build insists all three horns are rotated together — the radar only cares that transmit and receive agree.

Phase centre

The point the radiated field appears to come from. For a horn it sits inside the flare, not at the mouth, and it moves with frequency and differs between E- and H-planes. It does not matter for a one-antenna gain figure. It matters a great deal if you ever put two horns side by side and measure the phase difference between them to get a bearing, because that is the baseline you are dividing by.

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3 · Waveguide, because the horn is fed by one

A hollow metal pipe carries a wave only above a **cutoff frequency**. Below it the field decays exponentially — the guide is a high-pass filter, and that is a property of the geometry, not the material. Modes are named TE_{mn} and TM_{mn} for the number of half-wave variations across each dimension.

$$f_c(TE_{mn}) = (c/2) \cdot \sqrt{(m/a)^2 + (n/b)^2}$$

TE_{10} is the dominant mode — the lowest cutoff, one half-wave across the broad wall a and nothing across b . You want it alone: as soon as a second mode propagates, the aperture field is a sum of two patterns and the result is neither predictable nor stable. The single-mode window runs from $f_c(TE_{10}) = c/2a$ to $f_c(TE_{20}) = c/a$ — exactly one octave.

for WR-340 (86.4 × 43.2 mm)	value
TE_{10} cutoff, $c/2a$	1.735 GHz
TE_{20} cutoff, c/a	3.470 GHz
single-mode band	1.735 – 3.470 GHz — 2.45 GHz sits comfortably inside
guide wavelength λ_g at 2.45 GHz	173.3 mm (λ_0 is 122.4 mm)

$$\lambda_g = \lambda / \sqrt{1 - (\lambda/2a)^2} \text{ always longer than } \lambda$$

The guide wavelength is longer than free space, and it is λ_g — not λ — that sets every length inside the guide. Using $\lambda/4$ where $\lambda_g/4$ belongs is one of the most common ways a first waveguide design comes out mistuned: here it is the difference between 30.6 mm and 43.3 mm.

Getting energy in: the coax probe

A quarter-wave monopole pushed through the broad wall, parallel to the E-field, with a **short circuit $\lambda_g/4$ behind it**. The short reflects a wave that arrives back at the probe having travelled $\lambda_g/2$, in phase with the forward wave, so the two add going out of the guide and cancel going backward. Two knobs:

- **Probe depth** sets the coupling, and therefore the real part of the impedance. Too short under-couples; too deep starts to look capacitive.
- **Backshort distance** trims the reactance. $\lambda_g/4$ is the design value; a couple of millimetres either way is the fine adjustment.

For this horn: backshort at **43.3 mm**, probe about **29 mm** into a 43.2 mm guide. The probe must clear the far wall — check it, because a probe that touches is a short circuit and reads as a beautiful, entirely fictional S_{11} .

4 · Aperture antennas and horn design

Open the end of a waveguide and it radiates, badly: the sudden step from η_{guide} to 377Ω reflects a lot, and the aperture is small so the beam is broad. Flaring fixes both — a gradual taper matches better, and a larger mouth is a larger aperture. But flaring introduces the problem that horn theory exists to manage.

Aperture phase error — the central idea

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In a flared horn the wave spreads cylindrically/spherically from a virtual apex. The path from that apex to the edge of the mouth is longer than the path to the centre, so the field at the mouth is **not in phase across the aperture** — it lags at the edges. The error is quadratic, and it is expressed dimensionlessly as:

$$s = b_1^2 / (8 \cdot \lambda \cdot \rho_1) \quad (\text{E-plane}) \quad t = a_1^2 / (8 \cdot \lambda \cdot \rho_2) \quad (\text{H-plane})$$

Make the horn longer for a given mouth and ρ grows, the phase error falls, and the aperture efficiency rises toward 1 — but the horn gets absurd. Make it shorter and the efficiency collapses. The **optimum horn** is the shortest one for a given gain, and it sits at:

	optimum value	this horn
E-plane, s	$1/4 = 0.250$	0.250
H-plane, t	$3/8 = 0.375$	0.375
aperture efficiency	≈ 0.51	0.51

Those two fractions are the whole of optimum-horn design. Everything else — $b_1 = \sqrt{2\lambda\rho_1}$, $a_1 = \sqrt{3\lambda\rho_2}$, the 51 % efficiency — follows from them.

Three horns, one idea

type	flared in	use
H-plane sectoral	a only	fan beam, narrow in H
E-plane sectoral	b only	fan beam, narrow in E
Pyramidal	both	pencil beam — this build

The constraint that catches everyone

A pyramidal horn only closes as a solid if the E-plane and H-plane flares reach the aperture at the **same axial length**: $p_e = p_h$. Change a_1 or b_1 on its own and the geometry no longer meets — in CAD you get a shape that will not loft, and in HFSS you get a solid with a sliver in it that meshes badly and solves slowly. horn.py solves both together and prints the mismatch as a percentage for exactly this reason.

Beamwidth and gain, closed form

$$\text{HPBW}_E \approx 54 \cdot \lambda / b_1 \quad \text{HPBW}_H \approx 78 \cdot \lambda / a_1$$

The E-plane is narrower for the same aperture dimension because its illumination is uniform, while the H-plane inherits a cosine taper from the TE_{10} distribution. The taper is also why the H-plane sidelobes are lower. For this horn: **34° E, 36° H, 13.4 dBi**.

Note what a 34° beam means for a radar: at 5 m the beam is about 3 m across. The horn does almost no localising indoors. Any real bearing has to come from somewhere else — a scan, or the phase between two receivers.

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5 · Where the antenna meets the radar

Gain enters the radar equation **twice**, once on transmit and once on receive, so a decibel of antenna is worth two decibels of link:

$$P_r = P_t \cdot G_t \cdot G_r \cdot \lambda^2 \cdot \sigma / ((4\pi)^3 \cdot R^4)$$

- **R⁴, not R².** Doubling the range costs 12 dB, because the wave spreads on the way out and again on the way back.
- **λ² in the numerator** looks like lower frequency is better, but G for a fixed physical aperture goes as 1/λ², so two fixed apertures beat it back the other way.
- **σ, the radar cross-section**, is the wildest term: it is not the target's physical area, it varies enormously with aspect and frequency, and for a small drone it is a guess until measured.

Two antenna properties decide the rest of the radar's behaviour: **gain** sets how far, and **beamwidth** sets how well you can tell two targets apart in angle — and also, unhelpfully, how much of the room you are illuminating and therefore how much clutter comes back with the target.

Isolation, which is its own problem

A bistatic radar with separate TX and RX horns has the transmitter running continuously a few hundred millimetres from the receiver. The direct leakage is vastly stronger than any echo, and how much of it there is — S_{21} between the two horns — is an antenna question you can simulate. Here the target is $S_{21} \leq -35$ dB at 290 mm separation.

6 · Measuring it, and simulating it

On the bench

measurement	how	watch out for
S_{11}	One-port VNA sweep, calibrated at the connector.	Calibrate at the far end of the cable, not the VNA face.
Gain	Two identical antennas, known separation R, Friis: $G = (S_{21} + 20\log_{10}(4\pi R/\lambda))/2$.	R must exceed $2D^2/\lambda$. Ground reflections; get both antennas well off the floor.
Pattern	Rotate one antenna against a protractor, record S_{21} .	The far-field condition again, plus everything in the room reflecting.
Isolation	S_{21} between the two mounted horns.	Measure it in the final mount — the mount is part of the answer.

None of this needs an anechoic chamber to be useful. It needs you to know which number is being corrupted by the room, which is mostly the sidelobes and the deep nulls, not the main lobe or the match.

What a solver actually does

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method	solves	good at	used by
FEM	Volume meshed into tetrahedra, frequency by frequency.	Enclosed, inhomogeneous structures — waveguides, horns, connectors.	HFSS
MoM	Currents on surfaces, integral equation.	Open, mostly-metal radiators — wires, patches, large reflectors.	FEKO, NEC
FDTD	Time-stepping the grid, one run gives all frequencies.	Wideband and transient problems.	CST, Meep

FEM is the right tool for this horn because the interesting physics is inside a closed metal volume with a coax feed on it. The cost is that it solves one frequency at a time and interpolates between them.

Convergence is not accuracy

HFSS refines the mesh until the scattering parameters stop moving by more than ΔS between passes, then declares convergence. That means the solver has stopped changing its mind about the model you gave it. It says nothing about whether the model is the antenna you are going to build. A converged solution of a wrong geometry, a port that is too small, or an air box that is too close is converged and wrong. **Always check the answer against closed form** — that is what the numbers on the first page of the HFSS guide are for.

Ways to get a confidently wrong answer

- **Air box too close.** A radiation boundary nearer than about $\lambda/4$ reflects, and the pattern and S_{11} both change. $\lambda/4$ at 2.45 GHz is 30.6 mm.
- **Port too small or too large.** A coax wave port must enclose the dielectric and essentially nothing else. Too big and it starts supporting waveguide modes of its own.
- **Meshing the metal instead of the air.** What radiates is the field in the air volume. Model the air and give its walls a boundary; do not model 0.5 mm copper sheet and ask the solver to mesh it.
- **Gain above $4\pi A/\lambda^2$.** Physically impossible, so the model is wrong. Check first.
- **A pattern with no back lobe at all.** Real horns leak backwards. A perfectly clean back hemisphere usually means the radiation boundary is wrapping the structure rather than enclosing it.

7 · Symbols, quick reference

symbol	is	for this horn
λ	free-space wavelength	122.4 mm at 2.45 GHz
λ_g	guide wavelength, TE_{10}	173.3 mm
a, b	guide inside dimensions	86.4 × 43.2 mm (WR-340)
a_1 , b_1	aperture, H-plane and E-plane	263.8 × 193.1 mm
ρ_1 , ρ_2	E- and H-plane flare radii	152.4 / 189.6 mm

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symbol	is	for this horn
p_e, p_h	axial flare lengths (must be equal)	91.5 / 91.5 mm
s, t	aperture phase errors	0.250 / 0.375
e_{ap}	aperture efficiency	0.51
G	gain	13.4 dBi (bound 16.3)
η_0	free-space impedance	376.7 Ω
Γ, S_{11}	reflection at the feed	target ≤ -10 dB, 2.40–2.484 GHz

Work it out yourself

The fastest way to make any of this stick is to rederive the horn from λ alone. Try it: from $f = 2.45$ GHz get λ ; from WR-340's a get f_c and λ_g ; from $\rho_1 = 152.4$ mm get $b_1 = \sqrt{(2\lambda\rho_1)}$ and check it against 193.1 mm; then get the gain from $0.51 \cdot 4\pi a_1 b_1 / \lambda^2$ and the beamwidths from $54\lambda/b_1$ and $78\lambda/a_1$.

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